TrayGuard: What We Learned About Why Tray Errors Persist

EC ENGR 180DW — Final Design Review

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Roadmap

Introduction

The Problem By The Numbers

9 per 100 cases — Alfred et al. [1] 3,900 defects in 41,799 cases
55% at assembly

10 min avg delay — Nichol et al. [2] 236 errors, 147 cases
$6–9M/yr estimated cost

44% wrong spec — Zhu et al. [3] 398 errors in 33,839 packages
largest single category

These are recurring, measured operational failures — not isolated accidents.

The Problem is Deeper Than a Camera Can See

What Tray Reconstruction Actually Requires

FIGURE TBD
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- Identify each instrument by sight (no labels)
- Match against a count sheet with **local names**
- Distinguish lookalikes (straight vs curved Mayo)
- Check for damage, wear, contamination
- Count correct quantities
- All under **production pressure**

Six Root Causes

FIGURE TBD
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Three structural drivers:

1. SPD evolved from materials management → **no licensure, weak professionalization**
2. Cost-center accounting → wage investment is visible, error cost is invisible
3. No information architecture → downstream OR errors don't trigger upstream correction

The Broken Feedback Loop

FIGURE TBD
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"Error reporting is cumbersome, human-dependent, delayed, and incomplete." — Nichol et al.

We Evaluated Seven Solution Categories

Solution	Buildable?	Validatable?	Creates infra for future?
CV tray checker	No (needs SPD)	No	Maybe
Physical error-proofing	No (custom fab)	No	No
Workflow redesign	No (policy change)	No	No
Decision support	Partial	Partial	Yes (local content)
Task simplification	No (hospital data)	No	No
Accountability system	No (org policy)	No	No
Training (selected)	**Yes**	**Yes**	**Yes (module schema)**

Why Training, Not Because Training Is a Cure

What training addresses

- Measurable local tray knowledge
- Lookalike discrimination practice
- Pre/post learning evidence

What training does NOT fix

- Broken count sheets
- Staffing shortages
- Structural wage problems
- The broken feedback loop

Training is the one piece we can build and validate **now**.

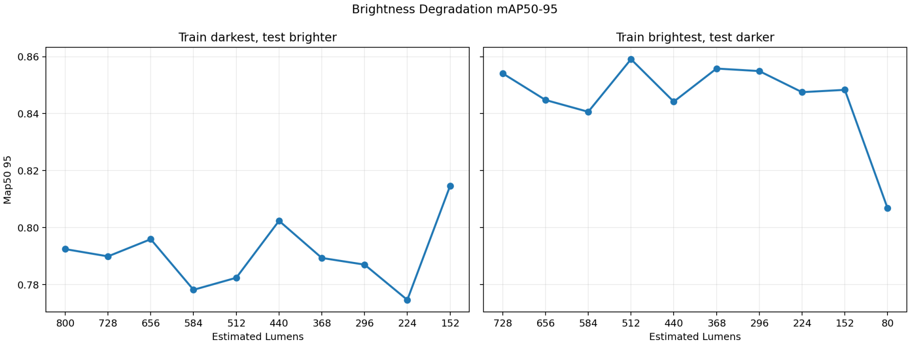
Why CV Alone Won't Work

We Ran 7 Staged Detection Experiments

Experiment	What it tested	Key result
Bright→dim	Light tolerance	mAP50-95 $=$ 0.781
Matte→reflective	Background transfer	mAP50-95 $=$ 0.759
Separated→overlay	Occlusion	**mAP50-95 = 0.389**
Real instruments	Shape vs position	Per-class AP: 0.000–0.862

Lavado baseline (clean data): **mAP50 = 0.990**. Pipeline works — the data doesn't generalize.

Brightness Degradation

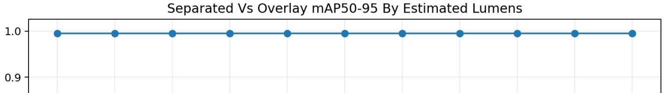


Occlusion Is the Dealbreaker

Separated layout (training)

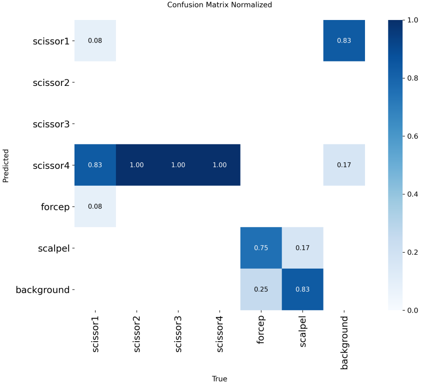
Overlay layout (test)

Overlay degradation

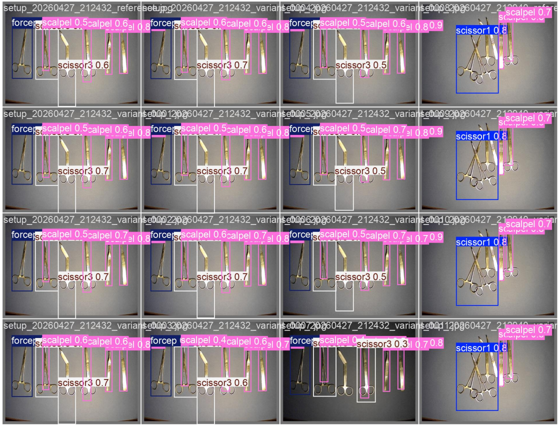


Real Instruments: The Model Learned Position, Not Shape

Confusion matrix



Prediction example



Split	Surviving classes	Zero AP classes
Matte→reflective	scissor2, forcep, scalpel	scissor1/3/4
Reflective→matte	**scalpel only**	scissor1/2/3/4, forcep

The Cumulative Lesson: CV Validation Burden

What a deployment-grade CV system still needs

- ✗ Site-specific training data (instruments, lighting, layouts)
- ✗ Cross-manufacturer validation [4] (Kienle: major performance drop)
- ✗ Open-set rejection (unknown objects = false positives)
- ✗ Human-in-the-loop for every uncertain case
- ✗ Workflow integration in live SPD (never tested)
- ✗ Regulatory / device policy / privacy review

The training path doesn't need any of this for a first study.

Why Training Is the Best Entry Point

CV and Training Are Complementary

FIGURE TBD
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The module is the durable artifact. The training loop proves the module works. The module then enables CV.

The Durable Design Artifact: Tray Module Schema

```
{
  "id": "mayo_scissors_straight_55",
  "display_name": "Straight Mayo, 5.5 in",
  "aliases": ["straight Mayo",
              "suture scissors"],
  "distinguishing_features": [
    "Heavy straight blades"
  ],
  "common_confusions": [
    "mayo_scissors_curved_55"
  ],
  "apriltag_id": 12,
  "study_prompts": [ ... ]
}
```

- Mirrors real count-sheet fields
- Adds aliases, lookalikes, prompts
- File-backed, versioned
- **Software is throwaway.**
- **Schema survives.**

Printed Cards: How It Works Physically

Front (assessment)

FIGURE TBD
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Back (study)

FIGURE TBD
figures/card˙back˙placeholder.pdf

Neutral front prevents answer leakage. Tag ID maps to instrument ID.

The Scoring Architecture: 5 Error Categories

Category	Definition	Example
Correct	Right item, right count	1 scalpel handle ✓
Missing	Required item not selected	No Kelly clamp
Wrong	Item not on tray list	Retractor not in this tray
Extra	Distractor selected	Item from different tray
Misidentified	Right class, wrong variant	Curved instead of straight
Wrong count	Right item, wrong quantity	2 Kellys, need 4

- Maps to literature error categories [1, 3, 2]
- Can separate *what* improved (fewer misses? fewer lookalikes?)

Confidence Capture

FIGURE TBD
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High-confidence error

Learner was sure but wrong. Targeted feedback needed.

Low-confidence correct

Learner was unsure but right. Fragile knowledge, needs reinforcement.

The Learning Loop (7 Steps)

FIGURE TBD
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Pre-test (baseline) → Study cards → Quiz → Practice sort → Post-test → Export

The Research Question

How can a local tray training and assessment system help novice users improve their ability to detect missing, wrong, extra, misidentified, or miscounted instruments before supervised SPD tray-reconstruction work?

That question is narrow, testable, and honest about what we can claim.

What a Future Team Should Do

The Study Ladder

FIGURE TBD
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Kirkpatrick Level 2 target. Level 3–4 require future validation.

Study 1: Novice Pre/Post Design

Segment	Time	Activity
Consent + intake	3–5 min	Background
Pre-test tray sort	5–8 min	No hints, no feedback
Retrieval cards	10–12 min	Prompt-before-reveal
Quiz	5–8 min	With confidence
Practice sort	8–10 min	Error feedback
Post-test	5–8 min	Parallel variant, no hints
Photo-ID test	5–7 min	Convergent validity

n = 8–12 novices, one 60-min session, within-subject

Study 2: Delayed Retention (The Falsification Criterion)

FIGURE TBD
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Criterion 1 — $\geq$ 50% of gain retained

Criterion 2 — Accuracy ¿ 5pp above baseline

Criterion 3 — $\geq$ 70% return rate

If these fail, the claim reverts to ”immediate learning only.”

Study 3: Expert Review (Closing the Content Gap)

- **Q** Are instrument names and aliases correct for local practice?
- **Q** Are counts and distractors plausible for training?
- **Q** Are lookalike pairs educationally meaningful?
- **Q** Are feedback messages misleading or safe?
- **Q** Would the weak-item report help target coaching?

The main weakness of a student-built module is **content validity**. Expert review closes that gap.

Go/No-Go Decision for Future Team

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Study passes → SPD pilot is justified. Study fails → problem may be harder than training alone.

Conclusion

What We Know and What We Don't

What we know

- Tray errors are systemic, recurring, and well-documented
- CV alone can't solve this — the validation burden is too high
- Training is the buildable entry point
- The module schema is durable infrastructure
- The study ladder has falsification criteria

What we can't claim (yet)

- Training transfers to real SPD work
- System works for experienced technicians
- It reduces OR delays
- CV is deployment-ready
- Same-day gains persist (Study 2 tests this)

The Honest Handoff

Our project found that this problem is harder and more systemic than any single technical fix can address. Our contribution is documenting *why*, and giving the next team a validated starting point — not a finished product.

The study is designed. The schema is defined. The scoring is tested. **Run the study.**

References I

[1] Alfred et al., "Work systems analysis of sterile processing: assembly," *BMJ Quality & Safety*, 2021. [Online]. Available: https://qualitysafety.bmj.com/content/30/4/271

[2] Nichol et al., "Observed rates of surgical instrument errors point to visualization tasks as being a critically vulnerable point in sterile processing and a significant cause of lost chargeable or minutes," *BMC Surgery*, 2024. [Online]. Available: https://link.springer.com/article/10.1186/s12893-024-02407-1

[3] Zhu et al., "Errors in packaging surgical instruments based on a surgical instrument tracking system: an observational study," *BMC Health Services Research*, 2019. [Online]. Available: https://link.springer.com/article/10.1016/j.arth.2022.08.017

[4] Kienle et al., "Surgical instrument detection on the instrument stand using neural networks," *Current Directions in Biomedical Engineering*, 2025. [Online]. Available: https://doi.org/10.1515/cdbme-2025-0235